

Assignment 10: Deploy a project from Github to EC2 by creating a new security group and user data.

Step 1: From EC2 page, click on Security groups.

The screenshot shows the AWS Management Console EC2 page. The left sidebar has a menu with 'EC2' selected, showing options like Dashboard, AWS Global View, Events, and Instances. The main content area is titled 'Resources' and lists various Amazon EC2 resources in the United States (N. Virginia) Region. The resources listed are: Instances (running) 0, Capacity Reservations 0, Elastic IPs 0, Key pairs 0, Placement groups 0, Snapshots 0, Auto Scaling Groups 0, Dedicated Hosts 0, Instances 0, Load balancers 0, Security groups 1, and Volumes 0. On the right, the 'EC2 cost' section shows the date range as 'Past 6 months', the region as 'Global', and the credits remaining as '\$139.96 USD'. It also shows the days remaining as 132 (July 7, 2026).

Step 2: Click on create security group.

The screenshot shows the AWS Management Console Security Groups page. The left sidebar has a menu with 'EC2' selected, showing options like Dashboard, AWS Global View, Events, and Instances. The main content area is titled 'Security Groups (1)' and shows a table with the following columns: Name, Security group ID, Security group name, and VPC ID. The table contains one entry: 'sg-08626cf47e94292f3' with the name 'default' and VPC ID 'vpc-0c8c38b5c5371ed1a'. There is a 'Create security group' button in the top right corner.

Step 3: Enter the required informations.

The screenshot shows the AWS Management Console 'Create security group' page. The page title is 'Create security group' and it includes a brief description: 'A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.' The 'Basic details' section contains the following fields: 'Security group name' (Practice1), 'Description' (test security group), and 'VPC' (vpc-0c8c38b5c5371ed1a).

Step 4: Set the Inbound rules as follows by clicking on Add rule.

The screenshot shows the AWS Management Console 'Inbound rules' page. The page title is 'Inbound rules' and it includes a brief description: 'A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.' The 'Inbound rules' section contains the following rules:

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	Any...	0.0.0.0/0
HTTP	TCP	80	Any...	0.0.0.0/0
HTTPS	TCP	443	Any...	0.0.0.0/0
Custom TCP	TCP	4000	Any...	0.0.0.0/0

There is an 'Add rule' button at the bottom left of the table.

Step 5: Click on Create security group.

The screenshot shows the 'Create security group' page in the AWS Management Console. The breadcrumb navigation is 'EC2 > Security Groups > Create security group'. The page has several sections: 'Outbound rules' with filters for Type (All traffic), Protocol (All), Port range (All), and Destination (0.0.0.0/0); a 'Tags - optional' section; and a 'Create security group' button. A warning message states: 'Rules with destination of 0.0.0.0/0 or :::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.'

Outbound rules Info

Type Info Protocol Info Port range Info Destination Info Description - optional Info

All traffic All All Cust... 0.0.0.0/0 X

Add rule

Rules with destination of 0.0.0.0/0 or :::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel Create security group

Step 6: Click on Launch instance from EC2 page.

The screenshot shows the 'EC2' page in the AWS Management Console. The breadcrumb navigation is 'EC2 > Resources'. The page displays a 'Resources' table with counts for various EC2 resources. A 'Launch instance' button is visible. The 'EC2 cost' section shows the current cost and credits remaining. The 'Service health' section indicates that the service is operating normally.

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Resources

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Resource	Count
Instances (running)	0
Dedicated Hosts	0
Key pairs	0
Security groups	1
Auto Scaling Groups	0
Elastic IPs	0
Load balancers	0
Snapshots	0
Capacity Reservations	0
Instances	0
Placement groups	0
Volumes	0

Launch instance

To get started, launch an Amazon EC2 Instance, which is a virtual server in the cloud.

Launch instance

Migrate a server

Note: Your instances will launch in the United States (N. Virginia) Region

Service health

AWS Health Dashboard

Region: United States (N. Virginia)

Status: This service is operating normally.

EC2 cost

Date range: Past 6 months

Region: Global

Costs in your free plan account are covered by credits.

Credits remaining: \$139.96 USD

Days remaining: 132 (July 7, 2026)

Cost (\$)

Activate Windows

Go to Settings to activate Windows.

Step 7: Give a name and set Ubuntu as the AMI.

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The breadcrumb navigation is 'EC2 > Instances > Launch an instance'. The page has several sections: 'Name and tags' with a text input for the instance name; 'Application and OS Images (Amazon Machine Image)' with a search bar and a 'Quick Start' section showing various AMIs; and a 'Summary' section on the right with a 'Launch instance' button.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

Practice

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances: 1

Software Image (AMI): Canonical, Ubuntu, 24.04, amd64...read more

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Activate Windows

Go to Settings to activate Windows.

Step 8: Select the existing security group and select the one just made.

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
prac2

Create new key pair

Network settings info

Network | Info
vpc-0c8c38b5c5371ed1a

Subnet | Info
No preference (Default subnet in any availability zone)

Auto-assign public IP | Info
Enable

Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

- ☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Summary

Number of instances | Info
1

Software image (AMI)
Canonical, Ubuntu, 24.04, amd64...read more
ami-0b6c6e6ed2801a5c5b

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel | Launch instance | Preview code

Step 9: Enter the following code as shown in the image and then click on Launch instance.

User data - optional | Info
Upload a file with your user data or enter it in the field.

Choose file

```
apt-get update
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -SL https://deb.nodesource.com/setup_16.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/Jbhawal/Cllg-Practice-AWS.git
cd Cllg-Practice-AWS
npm install
node index.js
```

☐ User data has already been base64 encoded

Summary

Number of instances | Info
1

Software image (AMI)
Canonical, Ubuntu, 24.04, amd64...read more
ami-0b6c6e6ed2801a5c5b

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel | Launch instance | Preview code

Step 10: Go to the instance page just created and copy the Public IPv4 address.

Instance summary for i-034e5c5fa1fc71490 | Info

Updated less than a minute ago

Instance ID
i-034e5c5fa1fc71490

IPv6 address
-

Hostname type
IP name: ip-172-31-24-31.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address

Public IPv4 address
100.31.156.195 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-24-31.ec2.internal

Instance type
t3.micro

VPC ID
vpc-0c8c38b5c5371ed1a

Private IPv4 addresses
172.31.24.31

Public DNS
ec2-100-31-156-195.compute-1.amazonaws.com | open address

Elastic IP addresses
-

AWS Compute Optimizer finding

Step 11: Paste it in a new tab' address bar followed by :4000 and we get the desired page running.

Launch an instance | EC2 | ap-south-1 | SecurityGroup | EC2 | ap-south-1 | Jbhawal/Cllg-Practice-AWS | 13.232.25.236:4000

Not secure 13.232.25.236:4000

Hello mckv